

# Quantum 2 HW3

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*Technion*

(Date: December 4, 2025)

## 1 Simple Model for Quantization of the Electromagnetic Field

1. Use Maxwell's equations to write time development equations for  $\mathcal{E}$  and  $\mathcal{B}$ .
2. Express  $U(t)$  using  $V, \mathcal{E}(t), \mathcal{B}(t), \mu_0, \varepsilon_0$ . Use

$$\int_V d\mathbf{r} \sin^2 kz = \int_V d\mathbf{r} \cos^2 kz = \frac{V}{2}$$

3. We'll define  $\omega = ck$  and

$$\chi(t) = \sqrt{\frac{\varepsilon_0 V}{2\hbar\omega}} \mathcal{E}(t) \qquad \Pi(t) = \sqrt{\frac{V}{2\mu_0\hbar\omega}} \mathcal{B}(t)$$

show that the equations of motion for  $\chi(t)$  and  $\Pi(t)$  are identical in their form to those of  $X$  and  $P$  in subsection 1.

### 1.1 Time Development

The relevant Maxwell's equations for time development are

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \qquad \nabla \times \mathbf{B} = \mu_0 \left( \mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \quad (1)$$

and since no current or charges at all other than out waves were specified, we assume the current  $\mathbf{J} = 0$ . Plugging the given fields into the equations we find

$$k\mathcal{E} = \frac{\partial \mathcal{B}}{\partial t} \qquad k\mathcal{B} = -\mu_0\varepsilon_0 \frac{\partial \mathcal{E}}{\partial t} \quad (2)$$

we can take the derivative of the equations and plug them into the other ones to find 2nd order differential equations

$$\frac{\partial^2 \mathcal{E}}{\partial t^2} + \frac{k^2}{\mu_0\varepsilon_0} \mathcal{E} = 0 \qquad \frac{\partial^2 \mathcal{B}}{\partial t^2} + \frac{k^2}{\mu_0\varepsilon_0} \mathcal{B} = 0 \quad (3)$$

## 1.2 Energy in Time

$$U(t) = \int_V d\mathbf{r} \left( \frac{\varepsilon_0}{2} \mathbf{E}^2 + \frac{1}{2\mu_0} \mathbf{B}^2 \right) \quad (4)$$

$$= \int_V d\mathbf{r} \left( \frac{\varepsilon_0}{2} \mathcal{E}^2 \sin^2(kz) + \frac{1}{2\mu_0} \mathcal{B}^2 \cos^2(kz) \right) \quad (5)$$

$$= \frac{V}{2} \left( \frac{\varepsilon_0}{2} \mathcal{E}^2 + \frac{1}{2\mu_0} \mathcal{B}^2 \right) \quad (6)$$

## 1.3 Same Form

The equations of motion for  $\mathcal{E}$  and  $\mathcal{B}$  are 2nd order linear ODEs, which are invariant to multiplication by scalars. The transformation of parameters from  $\mathcal{E}$  and  $\mathcal{B}$  to  $\chi$  and  $\Pi$  simply multiplies by a scalar, therefore the equations of motion will remain the same.

## 2 Quantization of the Field

1. Express  $\hat{H}_F$  in terms of  $\hat{a}$  and  $\hat{a}^\dagger$  and the number operator  $\hat{N} = \hat{a}^\dagger \hat{a}$ .
2. Calculate the expectation values  $\langle \hat{E}(\mathbf{r}) \rangle_n, \langle \hat{B}(\mathbf{r}) \rangle_n, \langle \hat{H}_F \rangle$  in state  $|n\rangle$ .

### 2.1 Field Hamiltonian

From the definitions for the photon creation and annihilation operators we can find expressions for  $\hat{\chi}$  and  $\hat{\Pi}$  using the operators:

$$\hat{\chi} = \frac{1}{\sqrt{2}} (\hat{a} + \hat{a}^\dagger) \quad \hat{\Pi} = \frac{1}{\sqrt{2}i} (\hat{a} - \hat{a}^\dagger) \quad (7)$$

therefore

$$\hat{H}_F = \frac{\hbar\omega}{2} (\hat{\chi}^2 + \hat{\Pi}^2) \quad (8)$$

$$= \frac{\hbar\omega}{4} ((\hat{a} + \hat{a}^\dagger)^2 - (\hat{a} - \hat{a}^\dagger)^2) \quad (9)$$

$$= \frac{\hbar\omega}{4} (\hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + \hat{a}^{\dagger 2} - \hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} - \hat{a}^{\dagger 2}) \quad (10)$$

$$= \frac{\hbar\omega}{2} (\hat{a}\hat{a}^\dagger + \hat{N}) \quad (11)$$

$$= \frac{\hbar\omega}{2} (\hat{a}^\dagger\hat{a} - \hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger + \hat{N}) \quad (12)$$

$$= \frac{\hbar\omega}{2} (2\hat{N} + [\hat{a}, \hat{a}^\dagger]) \quad (13)$$

$$= \frac{\hbar\omega}{2} (2\hat{N} + 1) \quad (14)$$

$$= \hbar\omega \left( \hat{N} + \frac{1}{2} \right) \quad (15)$$

### 2.2 Expectation Values

First we notice

$$\langle \hat{E}(\mathbf{r}) \rangle_n = \langle n | \hat{E} | n \rangle \quad (16)$$

which if we look at the definition of the creation and annihilation operators which define  $\hat{E}$ , the result of the above equation is always zero, therefore

$$\langle \hat{E}(\mathbf{r}) \rangle_n = 0 \quad (17)$$

and the same holds for  $\langle \hat{B}(\mathbf{r}) \rangle_n$ . For  $\hat{H}_F$ :

$$\langle n | \hat{H}_F | n \rangle = \hbar\omega \langle n | \left( \hat{N} + \frac{1}{2} \right) | n \rangle = \hbar\omega \left( n + \frac{1}{2} \right) \quad (18)$$

### 3 Coherent States of the Electromagnetic Field

The coherent states of the Electromagnetic field are defined as

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \quad \alpha \in \mathbb{C} \quad (19)$$

1. Show that the state  $|\alpha\rangle$  is a normalized eigenstate of the annihilation operator  $a$ . Find the corresponding eigenvalue. Calculate the expectation value  $\langle N \rangle_\alpha$  in this state.
2. Given  $|\psi(t=0)\rangle = |\alpha\rangle$ , show that

$$|\psi(t)\rangle = \exp\left(-i\frac{\omega t}{2}\right) |\alpha e^{-i\omega t}\rangle$$

3. Calculate the expectation values  $\langle \mathbf{E}(\mathbf{r}) \rangle_t$  and  $\langle \mathbf{B}(\mathbf{r}) \rangle_t$  in the state  $|\psi(t)\rangle$ . Assume  $\alpha$  is real.
4. Show that  $\langle \mathbf{E}(\mathbf{r}) \rangle_t$  and  $\langle \mathbf{B}(\mathbf{r}) \rangle_t$  satisfy Maxwell's equations.
5. Calculate the classical energy of the field for  $\mathbf{E}_{cl} = \langle \mathbf{E}(\mathbf{r}) \rangle_t$  and  $\mathbf{B}_{cl} = \langle \mathbf{B}(\mathbf{r}) \rangle_t$  and compare with  $\langle H_F \rangle_t$ .
6. Assume  $|\alpha| \gg 1$  and justify the statement that the states  $|\alpha\rangle$  have a "classical" character.

#### 3.1 Normalized Eigenstate

To show  $|\alpha\rangle$  is an eigenstate of the annihilation operator we'll activate it on the state:

$$a |\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \cdot a |n\rangle \quad (20)$$

$$= e^{-\frac{|\alpha|^2}{2}} \sum_{n=1}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \sqrt{n} |n-1\rangle \quad (21)$$

$$= e^{-\frac{|\alpha|^2}{2}} \sum_{n=1}^{\infty} \frac{\alpha^n}{\sqrt{(n-1)!}} |n-1\rangle \quad (22)$$

$$= e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^{n+1}}{\sqrt{n!}} |n\rangle \quad (23)$$

$$= \alpha e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \cdot a |n\rangle \quad (24)$$

$$= \alpha |\alpha\rangle \quad (25)$$

therefore  $|\alpha\rangle$  is an eigenstate of the annihilation operator with eigenvalue  $\alpha$ . In line 21 the sum begins from 1 because the ground state annihilates into a zero, so it can be removed from the sum. Next we check that

the state is normalized:

$$\langle \alpha | \alpha \rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \cdot e^{-\frac{|\alpha^*|^2}{2}} \sum_{n=0}^{\infty} \frac{(\alpha^*)^n}{\sqrt{n!}} \langle n | n \rangle \quad (26)$$

$$= e^{-|\alpha|^2} \sum_{n=0}^{\infty} \frac{\alpha^{*n} \cdot \alpha^n}{n!} \quad (27)$$

$$= e^{-|\alpha|^2} \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} \quad (28)$$

$$= e^{-|\alpha|^2} \cdot e^{|\alpha|^2} \quad (29)$$

$$= 1 \quad (30)$$

noting that in the general case it should be expanded as  $\langle n | m \rangle$  but since it obviously becomes  $\delta_{nm}$  I just skipped there directly. Next the expectation value of  $N$  is:

$$\langle N \rangle_{\alpha} = \langle \alpha | N | \alpha \rangle = \langle \alpha | a^{\dagger} a | \alpha \rangle = \alpha^* \alpha \quad (31)$$

### 3.2 Showing That

We know that  $|\alpha\rangle$  is an eigenstate of the annihilation operator with eigenvalue  $\alpha$ . As we know from question 2, the hamiltonian (which is time independent) of the EM field can be expressed using the creation and annihilation operators. This means the state  $|\alpha\rangle$  is also an eigenstate of the EM field, which means we can use standard time evolution of eigenstates inserting the energy from question 2:

$$|\psi(t)\rangle = e^{-i\omega t(N+\frac{1}{2})} |\alpha\rangle \quad (32)$$

$$= e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} e^{-i\omega t(n+\frac{1}{2})} \frac{\alpha^n}{\sqrt{n!}} |n\rangle \quad (33)$$

$$= e^{-\frac{|\alpha|^2}{2} - i\omega t} \sum_{n=0}^{\infty} \frac{(\alpha \cdot e^{-i\omega t})^n}{\sqrt{n!}} |n\rangle \quad (34)$$

$$= [\beta \equiv \alpha \cdot e^{-i\omega t}] = e^{-\frac{|\beta|^2}{2} - i\omega t} \sum_{n=0}^{\infty} \frac{\beta^n}{\sqrt{n!}} |n\rangle \quad (35)$$

so

$$|\psi(t)\rangle = e^{-\frac{i\omega t}{2}} |\beta\rangle = \exp\left(-i\frac{\omega t}{2}\right) |\alpha e^{-i\omega t}\rangle \quad (36)$$

### 3.3 Expectation Values

$$\langle \mathbf{E}(\mathbf{r}) \rangle_t = \langle \psi(t) | \mathbf{E} | \psi(t) \rangle \quad (37)$$

$$= \langle \alpha e^{i\omega t} | \mathbf{E} | \alpha e^{-i\omega t} \rangle \quad (38)$$

$$= \varepsilon_x \sqrt{\frac{\hbar\omega}{\varepsilon_0 V}} \sin(kz) \langle \alpha e^{i\omega t} | (\alpha^{\dagger} + \alpha) | \alpha e^{-i\omega t} \rangle \quad (39)$$

$$= \varepsilon_x \sqrt{\frac{\hbar\omega}{\varepsilon_0 V}} \sin(kz) (\alpha e^{-i\omega t} + \alpha e^{i\omega t}) \quad (40)$$

$$= 2\alpha\varepsilon_x \sqrt{\frac{\hbar\omega}{\varepsilon_0 V}} \sin(kz) \cos(\omega t) \quad (41)$$

and

$$\langle \mathbf{B}(\mathbf{r}) \rangle_t = \langle \alpha e^{i\omega t} | \mathbf{B} | \alpha e^{-i\omega t} \rangle \quad (42)$$

$$= i\varepsilon_y \sqrt{\frac{\mu_0 \hbar \omega}{V}} \cos(kz) \langle \alpha e^{i\omega t} | (a^\dagger - a) | \alpha e^{-i\omega t} \rangle \quad (43)$$

$$= i\varepsilon_y \sqrt{\frac{\mu_0 \hbar \omega}{V}} \cos(kz) (\alpha e^{i\omega t} - \alpha e^{-i\omega t}) \quad (44)$$

$$= -2\alpha\varepsilon_y \sqrt{\frac{\mu_0 \hbar \omega}{V}} \cos(kz) \sin(\omega t) \quad (45)$$

### 3.4 Satisfying Maxwell's Equations

For the first equation, on one hand

$$\nabla \times \langle E \rangle = \nabla \times \left( 2\alpha\varepsilon_x \sqrt{\frac{\hbar\omega}{\varepsilon_0 V}} \sin(kz) \cos(\omega t) \right) \quad (46)$$

$$= 2\alpha k \varepsilon_y \sqrt{\frac{\hbar\omega}{\varepsilon_0 V}} \cos(kz) \cos(\omega t) \quad (47)$$

$$= [\omega = ck, c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}] = 2\alpha ck \sqrt{\frac{\mu_0 \hbar ck}{V}} \cos(kz) \cos(ckt) \varepsilon_y \quad (48)$$

on the other hand

$$\frac{\partial \langle B \rangle}{\partial t} = -2\alpha\omega\varepsilon_y \sqrt{\frac{\mu_0 \hbar \omega}{V}} \cos(kz) \sin \cos \omega t \quad (49)$$

$$= -2\alpha ck \sqrt{\frac{\mu_0 \hbar ck}{V}} \cos(kz) \cos(ckt) \varepsilon_y \quad (50)$$

next

$$\nabla \times \langle B \rangle = \nabla \times \left( -2\alpha\varepsilon_y \sqrt{\frac{\mu_0 \hbar \omega}{V}} \cos(kz) \sin(\omega t) \right) \quad (51)$$

$$= -2\alpha k \varepsilon_x \sqrt{\frac{\mu_0 \hbar \omega}{V}} \sin(kz) \sin(\omega t) \quad (52)$$

$$= -2\alpha k \sqrt{\frac{\mu_0 \hbar ck}{V}} \sin(kz) \sin(ckt) \varepsilon_x \quad (53)$$

and

$$\mu_0 \varepsilon_0 \frac{\partial \langle E \rangle}{\partial t} = -2\alpha\omega\varepsilon_x \mu_0 \sqrt{\frac{\varepsilon_0 \hbar \omega}{V}} \sin(kz) \sin(\omega t) \quad (54)$$

$$= -2\alpha k \sqrt{\frac{\mu_0 \hbar ck}{V}} \sin(kz) \sin(ckt) \varepsilon_x \quad (55)$$

### 3.5 Classical Energy

The classical energy of the EM field is

$$U = \int_V d\mathbf{r} \left( \frac{\varepsilon_0}{2} \langle E \rangle^2 + \frac{1}{2\mu_0} \langle B \rangle^2 \right) \quad (56)$$

$$= \int_V d\mathbf{r} \left( \frac{\varepsilon_0}{2} \left( 2\alpha\varepsilon_x \sqrt{\frac{\hbar\omega}{\varepsilon_0 V}} \sin(kz) \cos(\omega t) \right)^2 + \frac{1}{2\mu_0} \left( -2\alpha\varepsilon_y \sqrt{\frac{\mu_0 \hbar\omega}{V}} \cos(kz) \sin(\omega t) \right)^2 \right) \quad (57)$$

$$= 2\alpha^2 \frac{\hbar\omega}{V} \int_V d\mathbf{r} (\sin^2(kz) \cos^2(\omega t) + \cos^2(kz) \sin^2(\omega t)) \quad (58)$$

$$= \alpha^2 \hbar\omega \quad (59)$$

In comparison, the energy from  $H_F$  is

$$\langle H_F \rangle = \hbar\omega \left( \langle \psi | N | \psi \rangle + \frac{1}{2} \right) = \hbar\omega \left( \alpha^2 + \frac{1}{2} \right) \quad (60)$$

So the classical energy is lower than the quantized energy.

### 3.6 Justification

If  $|\alpha| \gg 1$  then the classical energy and the quantized energy are effectively the same, the  $1/2$  term becoming too small to notice. In that case, the quantized behavior of the energy levels is effectively the same as the classical behavior.

## 4 Question 4

1. Show that the set of states  $A$  is an eigenbasis of  $H_0 = H_A + H_F$ . Find the eigenvalues.
2. Draw qualitatively on the energy axis the location of the five first eigenenergies of  $H_0$ . Show that aside from the ground state,  $H_0$ 's eigenstates are bunched in pairs of degenerate states.
3. The coupling between the field and the atom is a dipole coupling and is given by

$$W = \gamma (a\sigma_+ + a^\dagger\sigma_-)$$

where  $\gamma = -d \sin(kz_0) \sqrt{\hbar\omega/\varepsilon_0 V}$  where  $d$  is the dipole moment, determined experimentally.

- (a) Write the action of  $\hat{W}$  on the states  $|g, n\rangle$  and  $|e, n\rangle$
- (b) Which physical processes to the operators  $a\sigma_+$  and  $a^\dagger\sigma_-$  describe?